

AI/ML Engineering Interview Cheatsheet

20-30 page last 3-day revision guide

Prepared for Pune / Remote India AI-ML Engineering interviews

Basic + practical + production-grade knowledge in one structured set

How to use this PDF

Read for understanding first, then revise using the checklists and interview answer frames.

For each concept, practice saying: problem, data, model, evaluation, deployment, monitoring, failure handling.

This is written to reduce bookish answers and build production language for interviews.

3-Day Interview Revision Plan

Day	Focus	Output
Day 1	Classic ML, metrics, feature engineering, data leakage	Explain 20 concepts aloud.
Day 2	LLMs, RAG, LangGraph, fine-tuning, evaluation	Whiteboard 5 systems.
Day 3	Projects, system design, scenarios, HR answers	Do 2 full mock interviews.

1. How to answer any AI-ML question

- Define the concept simply.
- Explain where it is used.
- Give one production risk.
- Give one metric.
- Mention monitoring or rollback.

Answer flow



Say this in interviews

For how to answer any ai-ml question, I would not only explain the concept. I would define how I measure it, how I deploy it, how I monitor it and what I do when it fails.

Do say	Avoid saying
I would compare against a baseline and evaluate on representative examples.	I will just use the latest model.
I would track latency, cost, quality and failure rate after deployment.	Accuracy is high so it is done.
I would add fallback or human review for risky cases.	The LLM will handle it automatically.

2. Classic ML metrics

- Accuracy is weak for imbalance.
- Precision reduces false positives.
- Recall reduces false negatives.
- F1 balances precision and recall.
- PR-AUC is useful for rare positives.

Answer flow



Say this in interviews

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3. Feature engineering

- Start with business logic.
- Prevent leakage.
- Validate types and ranges.
- Version features.
- Monitor drift.

Answer flow



Say this in interviews

For feature engineering, I would not only explain the concept. I would define how I measure it, how I deploy it, how I monitor it and what I do when it fails.

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4. Model validation

- Use time split for time problems.
- Use group split if users repeat.
- Keep test set untouched.
- Compare to baseline.
- Do segment-level error analysis.

Answer flow



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5. Deep learning basics

- Weights learn patterns.
- Loss measures error.
- Backprop computes gradients.
- Regularization prevents overfit.
- Transformers use attention.

Answer flow



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6. LLM basics

- LLMs predict tokens.
- Context window is limited.
- Temperature changes randomness.
- LLMs can hallucinate.
- Structured output helps automation.

Answer flow



Say this in interviews

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7. RAG basics

- Parse documents.
- Chunk with metadata.
- Embed chunks.
- Retrieve relevant context.
- Generate grounded answer with citations.

Answer flow



Say this in interviews

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8. Advanced RAG

- Hybrid search improves recall.
- Reranking improves precision.
- Query rewriting handles vague questions.
- Corrective RAG retries weak retrieval.
- GraphRAG uses relationships.

Answer flow



Say this in interviews

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9. Agentic AI

- Agent has goal, state, tools and decisions.
- Use workflows for reliability.
- Limit tool permissions.
- Add human review.
- Trace every step.

Answer flow



Say this in interviews

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10. LangGraph

- Nodes are steps.
- Edges are transitions.
- State stores raw data.
- Conditional routing controls path.
- Checkpoints support recovery.

Answer flow



Say this in interviews

For langgraph, I would not only explain the concept. I would define how I measure it, how I deploy it, how I monitor it and what I do when it fails.

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11. Memory

- Short-term memory is current session state.
- Long-term memory persists across sessions.
- Vector memory retrieves semantically.
- Memory can be poisoned.
- PII needs consent and deletion.

Answer flow



Say this in interviews

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12. Fine-tuning

- Use RAG for new facts.
- Fine-tune for style or task behavior.
- Clean data matters most.
- LoRA trains adapters.
- Evaluate against baseline.

Answer flow



Say this in interviews

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13. LLM evaluation

- Build golden set.
- Score retrieval and generation separately.
- Track faithfulness and relevance.
- Use human review for calibration.
- Run regression tests before release.

Answer flow



Say this in interviews

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14. LLMOps

- Trace prompts and outputs.
- Track tokens and cost.
- Version prompts.
- Monitor latency.
- Collect feedback labels.

Answer flow



Say this in interviews

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15. MLOps

- Track experiments.
- Register models.
- Package with Docker.
- Deploy with CI/CD.
- Monitor drift and quality.

Answer flow



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16. Deployment

- API schema validation.
- Timeouts and retries.
- Autoscaling.
- Canary release.
- Rollback.

Answer flow



Say this in interviews

For deployment, I would not only explain the concept. I would define how I measure it, how I deploy it, how I monitor it and what I do when it fails.

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17. Local LLMs

- Use for privacy and cost.
- Quantization reduces memory.
- Small models need narrow tasks.
- Use routing.
- Do not expose local servers publicly.

Answer flow



Say this in interviews

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18. Security

- Treat user and retrieved text as untrusted.
- Use least privilege tools.
- Validate output.
- Redact PII.
- Log audit events.

Answer flow



Say this in interviews

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19. Cloud

- Use storage for raw data.
- Use containers for services.
- Use managed endpoints when needed.
- Monitor billing.
- Protect secrets.

Answer flow



Say this in interviews

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20. Vector DBs

- Store embedding plus metadata.
- Use filtering.
- Rebuild index after model change.
- Handle deleted docs.
- Evaluate recall@k.

Answer flow



Say this in interviews

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21. System design frame

- Clarify requirements.
- Define data flow.
- Choose baseline and model.
- Design APIs.
- Add observability and failure paths.

Answer flow



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22. Project explanation

- Problem.
- Data.
- Approach.
- Metrics.
- Deployment.
- Limitations.
- Next improvement.

Answer flow



Say this in interviews

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23. Scenario: RAG gives wrong answers

- Check parsing.
- Check chunking.
- Check embedding model.
- Check retrieval recall.
- Check prompt and citations.

Answer flow



Say this in interviews

For scenario: rag gives wrong answers, I would not only explain the concept. I would define how I measure it, how I deploy it, how I monitor it and what I do when it fails.

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24. Scenario: model degrades in production

- Check data drift.
- Check label delay.
- Check pipeline changes.
- Compare old/new predictions.
- Rollback if user impact is high.

Answer flow



Say this in interviews

For scenario: model degrades in production, I would not only explain the concept. I would define how I measure it, how I deploy it, how I monitor it and what I do when it fails.

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25. Final mental checklist

- Metric?
- Data?
- Baseline?
- Evaluation?
- Deployment?
- Monitoring?
- Security?
- Fallback?
- Rollback?
- Business impact?

Answer flow



Say this in interviews

For final mental checklist, I would not only explain the concept. I would define how I measure it, how I deploy it, how I monitor it and what I do when it fails.

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